

Packager Guidelines

Section 4: Compressor Cooling

4-1. Frame Oil Cooler

All HSRC compressors come standard with a factory supplied frame oil cooler to remove heat generated by the friction of moving parts. It is shipped loose from the compressor and is the responsibility of the packager to incorporate it into the package. The size and type of cooler largely depends on the amount of heat required to remove from the oil that absorbs this heat; this is typically referred to as "heat rejection", or "heat load". The frame oil heat rejection requirements vary mainly with compressor size, speed, and ambient conditions. The maximum heat load for each compressor model are listed in Table 3; these loads are based on the most severe anticipated service conditions (maximum speed). The factory supplied oil coolers are sized for this condition with the minimum required cooling water flow rates listed in Table 3 as well; note that these flow rates are based on using water with a 140°F (60°C) inlet temperature. The cooling water temperature to the cooler under any circumstance should never be above 150°F (86°C). To provide for the maximum service interval, the cooling water supplied to this cooler should be clean, with an antifreeze additive for operation in low ambient temperature. The cooler water flow should be regulated so that the oil temperature is held to a minimum of 130°F (54°C) to prevent excessive oil pressure drops and oil sludging in the crankcase.

For applications where a factory supplied oil cooler is not furnished and the application involves speeds lower than the maximum, a direct ratio of the speeds (actual/maximum) can be used as a multiplier to determine the required (lower) heat load. Example: A 6HOS2 operating at 890 rpm will have a maximum heat load of $(890/1200) \cdot 800 = 593$ BTU/min.

CAUTION

It may be necessary to provide auxiliary heating to bring the inlet water temperature up to the required minimum, or to provide auxiliary cooling to lower the inlet water temperature to acceptable levels.

Table 3. Frame Heat Rejection & Cooling Water Requirements (Oil Cooler)

Model	Max. Speed (RPM)	Max Heat Load (BTU/Minute)	Water Flow GPM (LPM)	Oil Pump Cap. GPM (LPM)	Oil to Cooler & Filter GPM (LPM)	Model	Max. Speed (RPM)	Max Heat Load (BTU/Minute)	Water Flow GPM (LPM)	Oil Pump Cap. GPM (LPM)	Oil to Cooler & Filter GPM (LPM)
5MOS2	1500	800	20 (76)	32 (121)	16 (61)	5HOS2	1500	900	20 (76)	40 (151)	23 (87)
5MOS4	1500	1600	30 (144)	71 (269)	32 (121)	5HOS4	1500	1800	30 (114)	70 (265)	46 (114)
5MOS6	1500	2400	40 (151)	97 (367)	48 (182)	5HOS6	1500	2700	40 (151)	110 (416)	69 (261)
6MOS2	1200	700	20 (76)	31 (117)	16 (61)	6HOS2	1200	800	20 (76)	40 (151)	23 (87)
6MOS4	1200	1400	30 (114)	73 (276)	32 (121)	6HOS4	1200	1600	30 (114)	70 (265)	46 (114)
6MOS6	1200	2100	40 (151)	99 (374)	48 (182)	6HOS6	1200	2400	40 (151)	110 (416)	69 (261)
7MOS2	1000	600	20 (76)	32 (121)	16 (61)	7HOS2	1000	700	20 (76)	40 (151)	23 (87)
7MOS4	1000	1200	30 (114)	73 (276)	32 (121)	7HOS4	1000	1400	30 (114)	70 (265)	46 (114)
7MOS6	1000	1800	40 (151)	98 (371)	48 (182)	7HOS6	1000	2100	40 (151)	110 (416)	69 (261)
						5HOSS2	1500	1100	20 (76)	63 (238)	32 (121)
						5HOSS4	1500	2200	30 (114)	94 (356)	64 (242)
						5HOSS6	1500	3300	40 (151)	125 (473)	96 (363)
						6HOSS2	1200	1000	20 (76)	63 (238)	32 (121)
						6HOSS4	1200	2000	30 (114)	94 (356)	64 (242)
						6HOSS6	1200	3000	40 (151)	125 (473)	96 (363)
						7HOSS2	1000	900	20 (76)	63 (238)	32 (121)
						7HOSS4	1000	1800	30 (114)	94 (356)	64 (242)
						7HOSS6	1000	2700	40 (151)	125 (473)	96 (363)

If an air-cooled oil cooler is installed in lieu of the standard water-cooled oil cooler, a thermostatic valve should be installed in the oil system to maintain the oil at or above the minimum temperature.

4-1.1. Thermostatic Control Valves

Temperature Settings

Pre-set Thermostatic Valves, set to predetermined temperatures at the factory, eliminates costly errors due to mistakes of operating personnel. Operators do not have the capability of changing operating temperatures, running equipment hot or cold. One pre-set Thermostatic Valve must be replaced with another one to make such a change. Siemens Energy typically recommends a Thermostatic Valve be preset to 160°F (71°C).

To extend the life of Thermostatic Valves, they should not be run for extended periods above 25°F (14°C) over their normal ratings. If temperature requirements are greater, contact the factory for recommendations.

Operations

The operating force is generated by the expansion of a special wax which reacts to very small changes in temperature.

An element assembly of the Model B series with the sliding valve in the cold position is shown in Figure 1A. The wax like fluid travels out the bypass (Port B as marked on the valve), the direction indicated by the arrow.

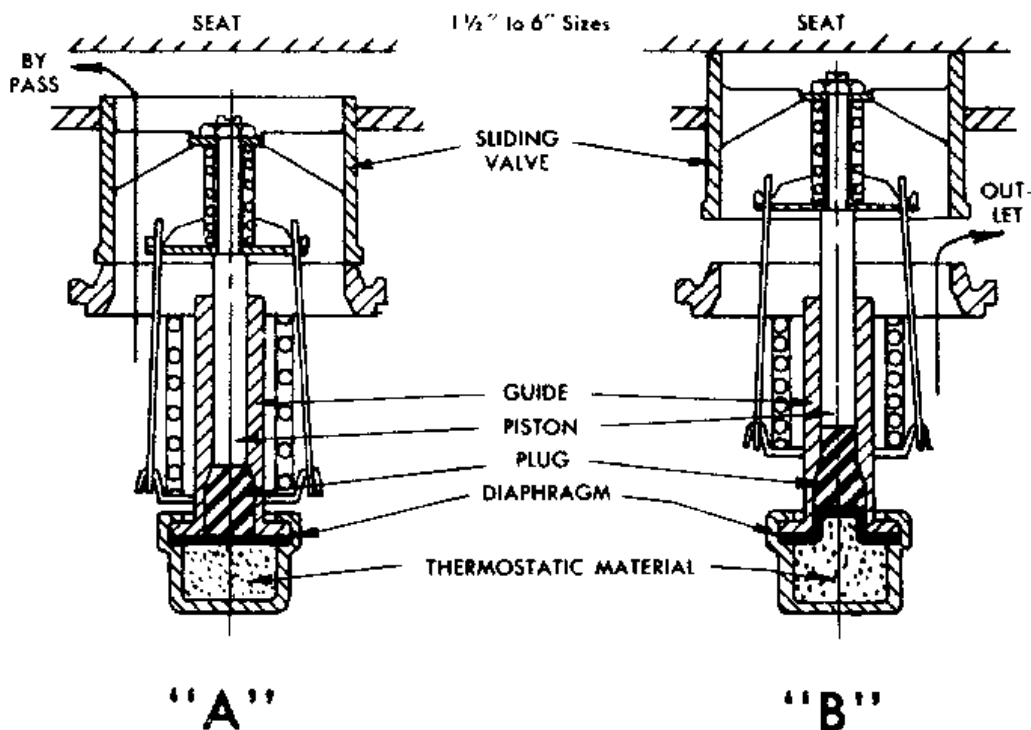


Fig. 1.

The sliding valve moves up to the warm, extended position as shown in Figure 1B. The Bypass is closed off by the sliding valve as it seats and the water is diverted to the outlet (Port C on the valve) as indicated by the arrow head at the Outlet.

While operating, the sliding valve is usually in the mid-position. Rising temperature expands the wax material, forcing the rubber plug into a reduced diameter in the piston guide, which multiplies the piston's movement by this extruding action.

The chemical composition of the wax material determines the operating range of the valve. The expansion of the wax generates pressure, transferring it to the piston, overcoming a spring force of greater than 100 lbs. (45.4 kg).

A simple, rugged construction is sensitive to changes in temperature. Valve construction nullifies the affects of pressure on the element. Surges in pressure do not tend to upset the stability of the thermostatic valve.

Piping Diagrams

Figure 2 represents the most common piping diagram for an engine jacket water temperature control. A radiator may replace a heat exchanger. The Thermostatic Valve will operate in any position and may be positioned with regard to convenience only.

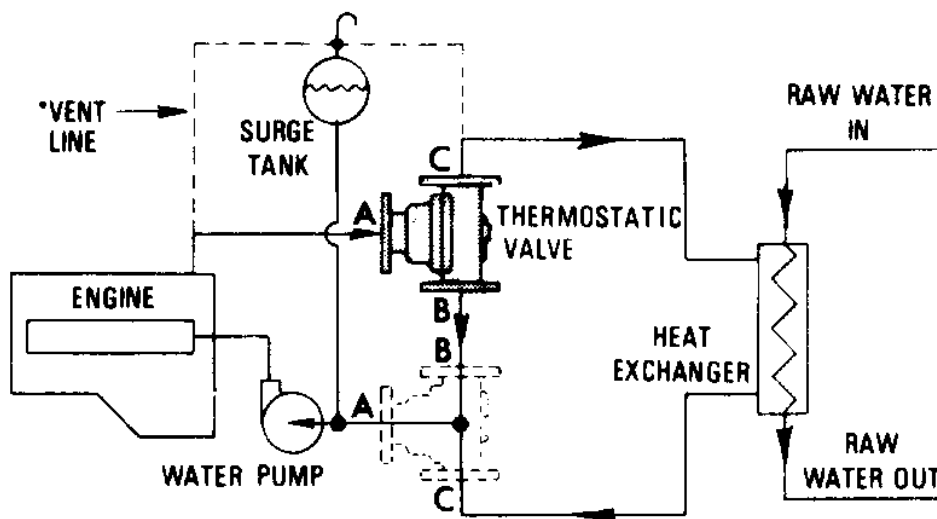


Fig. 2.

Figure 2. Cooling water control – Heat Exchanger Valve shown in “diverting” installation. Mount valve in dotted position for “mixing” applications.

Lubricating oil temperatures control for compressors utilizes the thermostatic valve installed directly in the lubricating oil line as illustrated in Figure 3. The oil bypasses the heat exchanger when cold will reach the desired operating temperature rapidly. When warmer, the correct amount of oil will automatically be circulated through the heat exchanger to maintain the desire temperature.

The system illustrated in Figure 3A is similar to the standard jacket water temperature control system with the thermostatic valve used as a diverting valve. In Figure 3B the Thermostatic Valve is used as a mixing valve, in which hot oil enters Port B and cold oil enters Port C. The oil is proportionally mixed so as to exit from Port A at the appropriate temperature.

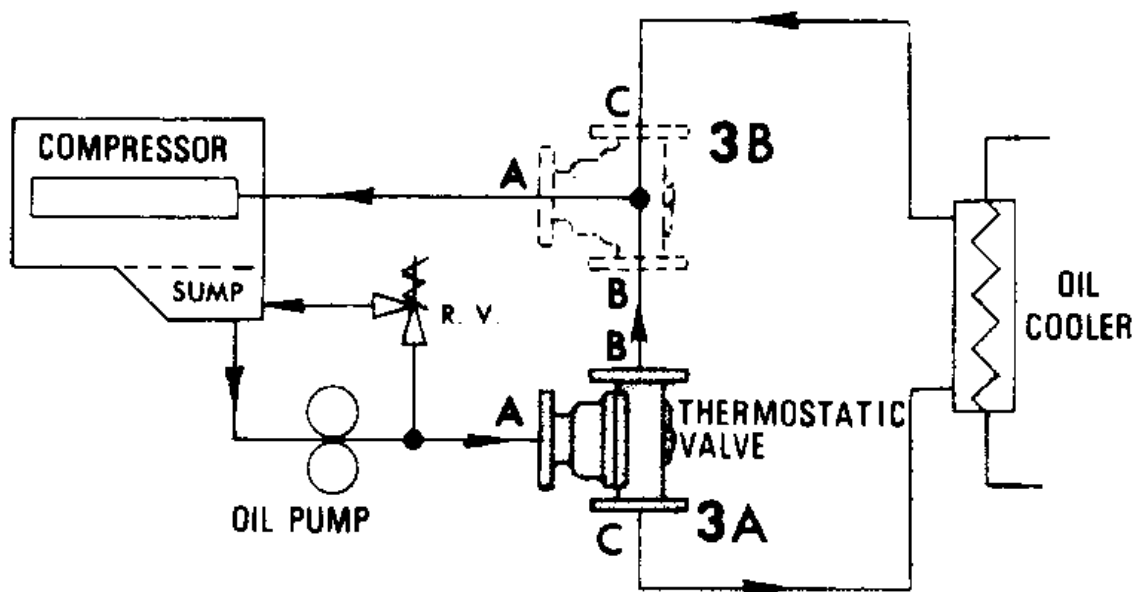


Fig. 3.

Figure 3. Lube Oil Control Valve shown in diverting position to control oil sump temperature. In dotted position valve will “mix” hot and cold flow streams to control supply temperature to compressor frame.

4-2. Cylinder Cooling

Siemens Energy Gas Field Compressor Cylinders - both gas cooled and water jacketed - are designed for 350°F (177°C) maximum temperature. Our Siemens Energy Sizing Performance Program has warning flags basis:

Forced cooling is required when the following is met or exceeded...

Discharge Temperature, Adiabatic	300°F (149°C)
Discharge Temperature, Estimated Actual	350°F (177°C)
Cylinder Average Temperature $((T_i + T_d, \text{adiabatic})/2)$	220°F (104°C)

Water jacketed cylinders may be applied with or without circulating coolant. If applied without circulating coolant, the water jackets must be vented to atmosphere. If they can be filled with coolant (normally water/glycol mix) and vented with provision for coolant expansion/contraction such as a standpipe or expansion tank above the cylinders, this would be ideal, but is not required.

Individual consideration should be given to each compressor application when selecting a particular cooling method. Cylinder design temperature should never be exceeded under any condition, but the User or Packager has the option to use a circulating water cooling system when jacketed cylinders are supplied.

4-2.1. Circulated Water Cooling

Circulating coolant through cylinder water jackets is used to remove some of the heat of compression and frictional heat from the cylinders and to provide uniform cylinder wall temperatures. Circulating systems are most often used when gas discharge temperatures approach or exceed 300°F (149°C) but may be used at lower temperatures if desired --- or when handling special gases that have problems at high temperatures such as H₂, etc.

All forged steel cylinders require a circulating jacket water system and water cooled packing system. See Paragraph 4.3.1 for other cylinders requiring water cooled packing systems.

Coolant flow rate will vary depending upon the horsepower required by the cylinder and the temperature rise of the coolant.

Skid-mounted units are usually provided with a closed loop cooling system, using a water/glycol mix for coolant. A surge tank should be included to allow for thermal expansion and makeup. A provision should be made to vent the high point in the system (usually the aerial cooler) through the surge tank. When filling the cooling system prior to starting the unit, it is important that all air is removed from the cylinder passages and jackets to assure complete and uniform cooling of the cylinder during operation.

CAUTION

To prevent condensation from forming on the cylinder walls and gas passages on all cylinders compressing wet gas (with water vapor present), the water inlet temperature to the cylinder should be at least 10°F (5.6°C) above the gas suction temperature.

If the water inlet temperature is above the gas inlet temperature to the cylinder, no condensate will form in most cases. Should condensate form on the cylinder walls and gas passages, it will tend to break down the lubrication qualities of oil and cause excessive wear to the cylinder bore, piston rings and valves, and possibly result in valve breakage.

The minimum recommended water outlet temperature from the cylinder jackets is about 110°F (45°C). This will result in optimum oil flow and distribution on the cylinder walls. In all cases, adequate flow must be maintained with a temperature rise across the cylinder of between 10°F (5.6°C) and 20°F (11° C).

4-3. Packing Coolant

4.3.1 Water Cooled

Some packing assemblies require cooling water to remove the heat of friction generated within the case. This is required on high pressure applications, long packing cases or boxes, and/or when there are non-lube requirements (see Table 4). The factors affecting the generation of heat are so varied and differ from application to application that an accurate prediction of the heat duty becomes very difficult. The best that can be done is to apply a general method of calculation, make certain assumptions, and then correct as necessary based on field result. Table 5 provides an estimate of the coolant flow and respective heat duty of the packing in normal service. It assumes a 140°F (60°C) coolant inlet temperature.

The coolant from the auxiliary coolant pump on the engine upstream of the thermostat on an engine driven unit can be used as long as the maximum temperature requirements are adhered to and minimum pressure is no lower than 25 PSIG (1.76 kg/cm²). These temperatures and pressures need to be checked before installing the system on the entire range of speeds that the engine will be operating. All of the tubing needs to be parallel, not in series; otherwise the head pressure will be too high. There also needs to be a flow indicator at each point to ensure the flow is adequate. If you have any questions regarding the setup of a cooling system, please contact the Siemens Energy Marketing or Engineering Department.

Table 4. Forced Cooling Required

Type Cylinder	Normal Operating Discharge Pressure PSIG (kg/cm ²)
MOS, HOS, HOSS	1750 (123) or Higher
All (Non-Lube)	250 (17.6) or Higher

NOTE

Due to the excessive pressure drop associated with internally cooled packings, it is recommended that the packing cooling system/piping be independent from any other water system. An inlet coolant pressure to the packing case of 35 PSIG (2.46 kg/cm²) is recommended along with a minimum pressure drop of 15 PSIG (1.05 kg/cm²). The coolant must be clean, filtered, softened and treated to prevent corrosion or plugging of the water passages in the packing cups.

Table 5. Cooling Requirements (per packing)

Type Machine	FLOW GPM (LPM)	HEAT LOAD BTU/Minute (joules/minute)
HOSS	2.9 (11.0)	244 (257433.6)
HOS	2.5 (9.5)	210 (221561.7)
MOS	2.3 (8.7)	193 (203836.8)

NOTE

Any cylinder that can be potentially operated at the above listed pressures will be equipped with water cooled cases; this does not mean a water cooling system is required to feed the particular case. Once again, it's pressure dependent, not cylinder dependent.

4-4. Precautions

- The coolant must be clean, softened and treated to prevent corrosion or plugging of the water passages in the packing cups.
- Do not exceed pressure or temperature design limitations.
- Do not allow water to freeze in the piping or cooler; use antifreeze.
- Turn on the cooling water before starting the unit.
- For closed system, it is recommended using a 50/50 solution of deionized water and glycol.